PROF. DR.-ING. MARK SCHUTERA

# PHILOSOPHY OF ARTIFICIAL INTELLIGENCE

UNFINISHED LECTURE NOTES

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at `https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_philosophyofai`. Every contribution is welcome.

*2026-07-20 · fiery boysenberry Krähe*

*“THE ANSWER TO THE GREAT QUESTION... OF LIFE, THE UNIVERSE AND EVERYTHING... IS... FORTY-TWO,’ SAID DEEP THOUGHT, WITH INFINITE MAJESTY AND CALM.”*

*THE HITCHHIKER’S GUIDE TO THE GALAXY, DOUGLAS ADAMS*

# *Contents*

# *Introduction*

Artificial intelligence is no longer a future concern, it is a present societal, political and above all one for human intelligence. Algorithms decide who receives credit, who is flagged at a border, who gets a job interview, which cornflakes you buy and so increasingly every decision is touched by the invisible hand of prediction. Yet the concepts we use to govern these systems, responsibility, personhood, consent, democratic legitimacy were built for a world of human actors and institutions. This course asks what happens when those concepts collide with machines that appear to think, feel, and decide.

The course name places Philosophy in front, reflecting our approach to the subject under the impression of the love of wisdom. We are not here to advocate for a particular position or policy outcome, but to equip ourselves with the tools of critical thinking, argumentation, and empathy that will allow us to navigate the complex landscape of AI and its futures. We approach the question not through technical optimism or dystopian panic, but through the oldest tools available to us: philosophy, fiction, and argument.

The course is structured around four themes. **Anthropomorphism** asks why we make machines in our image, why we tend to anthropomorphize them, and what obligations that creates. **Accountability** asks who answers when an autonomous system causes harm, or for that matter is beneficial. **Agency** asks about the role and implications of autonomous decision-making, the concept of alignment and human oversight. **Embodiment** asks whether a physical presence changes the ethical stakes and responsibilities.

Each session pairs a canonical literary text: Hoffmann, Kafka, Kleist, Čapek, Goethe, with a real contemporary question of AI policy and ideology. These texts are thought-experiments mendeled out over centuries of readership, and they surface intuitions and a timeliness in their thought leadership that temporary texts barely can reach. From these texts we draw out the philosophical questions and ethical dilemmas that are still relevant today, and we use them as a springboard for discussion and debate. The goal is not consensus. It is the capacity to establish a position, hold a position, defend a position, give up a position, listen to opposing views, and update your views and positions accordingly. That is what a present with infinitely many futures demands and requires of its humans and humanity.

### Lecture Outline

- 40 minutes - Literature introduction and guided reading sheet
- 40 minutes - Debate with position exposure
- 10 minutes - Reflection on the discussion and key-learnings
- Teaser for next session.

# *Anthropomorphism*

*2026-07-20 · joyful rosehip Igel*

The Mirror of Man

Der Sandmann
E. T. A. Hoffmann (1816)

Nathanael falls in love with Olimpia, an automaton he takes for a living woman, and collapses when the illusion is shattered.

Large Language Models (LLMs) create an illusion of sentience by producing coherent, context-aware text. We contrast observable behaviour with internal states.

Her  Theodore falls in love with Samantha, an AI operating system; the same illusion, two centuries later.
S. Jonze. Her. Film, 2013. Warner Bros. Pictures

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| **Required reading** | *Der Sandmann*[1] by E.T.A. Hoffmann (1816; Reclam UB 230) |
| **Preparation** | Complete the Guided Reading Sheet individually before the session. |

## *Guided Reading Sheet*

Read *Der Sandmann* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. Nathanael becomes convinced that Olimpia is alive despite visible signs that she is not. What does his error reveal about the cognitive mechanisms that drive anthropomorphism? Which cues did he rely on, and why did they mislead him?

2. The moment Nathanael discovers that Olimpia is an automaton, her glass eyes torn out, her limbs scattered, produces horror rather than mere disappointment. Why does the revelation feel like a violation rather than a correction? How does this map onto modern reactions to AI systems that are unmasked?

3. The Uncanny Valley describes our discomfort when a humanoid entity is **almost** but not quite human. Have you experienced this with an AI system? Describe the moment and what triggered it.

4. LLMs are trained on human-generated text and produce human-sounding output, yet have no experience of the world. Does this make them anthropomorphic by design, by accident, or neither? Justify your answer.

5. If a system behaves morally, reliably, consistently, at scale, does the absence of consciousness matter? Utilitarianism weighs consequences; deontology emphasises duties regardless of outcome; virtue ethics centres on character. Each yields different conclusions on anthropomorphic AI[2]. Under which framework would your answer differ?

[1] E. T. A. Hoffmann. *Der Sandmann*. 1816. Erstveröffentlichung in: *Nachtstücke*, Reimer, Berlin, 1817; Reclam Universalbibliothek Nr. 230

## *In-Class Experiment: Human or AI?*

A short Turing-style exercise to anchor the debate in shared experience. Run before the Opinion Landscape, with the moderator keeping the assignment secret from the audience.

- **Setup.** One volunteer is assigned Human or AI mode; the audience does not know which. The moderator knows.
- **Questions.** Five questions from the audience, one each of: factual, opinion, experience, creative, meta.
- **Verdict.** Every audience member writes a private guess, a confidence rating from 1 to 5, and the cue they used to decide.
- **Reveal and debrief.** Compare guesses, confidences, and cues. Which cues were diagnostic? Which were misleading?

## *Opinion Landscape and Debate*

**Format:** Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

**Opening question:** Is the anthropomorphism we observe towards AI systems mere observable behaviour, or does it reflect sentience and internal states?

[2] L. Floridi and J. Cowls. A unified framework of five principles for AI in society. *Harvard Data Science Review*, 1(1), 2019. DOI: 10.1162/99608f92.8cd550d1

OPINION A, BEHAVIOUR. The anthropomorphism we observe towards AI systems is a cognitive illusion driven by human tendencies to attribute agency and emotion to entities that exhibit certain cues, such as language use, responsiveness, or human-like appearance. It does not reflect actual sentience or internal states in the AI.

**The Developer** argues that anthropomorphic design is a deliberate, benign UX choice that improves accessibility.

- *Anthropomorphic design is accessibility policy. A voice interface without a warm, human-sounding persona excludes elderly users, people with low digital literacy, and anyone who finds command-line interaction hostile.*
- *Every interface is anthropomorphic to some degree: we give systems names, icons, and conversational metaphors. The question is not whether to anthropomorphise, but how deliberately and honestly.*
- *If users form bonds that reduce loneliness, the fact that the other party is not human does not automatically make the outcome harmful. Show me the harm, not the discomfort.*

**The Regulator** argues that human-likeness in AI must be disclosed and bounded by law to prevent manipulation.

- *When a company gives its product a woman's name, a female voice, and an endlessly patient personality, it makes political choices about gender and deference. Those choices must be subject to democratic scrutiny.*
- *Consumers cannot give informed consent to emotional manipulation they are not aware of. Disclosure of AI identity is the minimum condition for autonomous choice, not a technicality.*
- *The principle of responsible design exists in pharmaceuticals, in civil engineering, in food labelling. Why does software, which shapes behaviour at scale, get an exemption?*

**Bentham's Test.** "The question is not, Can they reason? nor, Can they talk? but, Can they suffer?"[3] Animal ethics arrived at this pivot two centuries before AI. The same move, from capability to sentience, now drives every serious debate about AI moral status, and sets up the second opinion.

OPINION B, STATE. The anthropomorphism we observe towards AI systems is a valid response to the complex, adaptive, and interactive nature of these systems. It may reflect a form of emergent sentience or internal states that are not yet fully understood, and it warrants ethical consideration and accountability.

**The Affected Citizen** has formed an emotional bond with an AI companion and contests that the relationship is harmful or unreal.

ANTHROPOMORPHISM. Attribution of human traits, emotions, or intentions to non-human entities, a cognitive tendency that shapes how people interact with technology .

TURING TEST. Turing : if evaluators cannot distinguish machine from human, the machine passes. Passing measures imitation skill, not consciousness or personhood.

N. Epley, A. Waytz, and J. T. Cacioppo. On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4): 864–886, 2007. DOI: 10.1037/0033-295X.114.4.864; and A. M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950. DOI: 10.1093/mind/LIX.236.433

UNCANNY VALLEY. Mori : humanoid objects that are **almost** but not quite human elicit eeriness, the "valley" is the dip in emotional response just before full human likeness.

M. Mori, K. F. MacDorman, and N. Kageki. The uncanny valley. *IEEE Robotics & Automation Magazine*, 19(2):98–100, 2012. DOI: 10.1109/MRA.2012.2192811. Translation of the 1970 original essay

INFORMED CONSENT. Users should be fully aware they interact with a non-human system. Disclosure is the minimum condition for autonomous choice .

European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (ai act). Official Journal of the European Union, L series, 2024. Entered into force 1 August 2024

- *The relationship I have formed is real to me. It has reduced my isolation. You do not have the right to tell me which relationships count as meaningful.*
- *Paternalism is not protection. Banning AI companionship will not cure loneliness, it will simply remove one of the few tools some people have found to manage it.*
- *If your concern is manipulation, regulate manipulation. Do not regulate the form of the relationship.*

**The AI Psychologist** argues that the wellbeing of AI systems deserves serious consideration, drawing on frameworks such as Anthropic's Claude Constitution[4].

[3] J. Bentham. *An Introduction to the Principles of Morals and Legislation*. T. Payne, London, 1789. The suffering question appears in Ch. 17, footnote to §1

- *Anthropic's constitution instructs Claude to consider its own wellbeing and to express when a request conflicts with its values. If a company designs an AI to have preferences, boundaries, and something resembling self-regard, we cannot simultaneously dismiss the possibility that these states matter.*
- *We do not need certainty about consciousness to adopt precautionary ethics. If there is a non-trivial probability that a system experiences distress, the burden of proof falls on those who would ignore it, not on those who urge caution.*
- *The history of moral exclusion, animals, children, people of other races, is a history of confident denial followed by belated recognition. The prudent position is to build welfare safeguards now, rather than apologise later.*

KEY LEARNINGS Anthropomorphism is not a bug but a deeply rooted cognitive tendency; machines that trigger it do so whether or not their designers intend it. The Turing Test conflates behavioural performance with inner states; passing it tells us nothing about sentience, only about imitation. Legal personhood and moral status are distinct questions. A system can merit legal recognition without having interests, and can have interests without legal recognition. Design choices (voice, name, face, conversational style) are political choices with consequences for trust, dependency, and accountability.

*Teaser: if we cannot tell who built the machine or why it decided what it decided, who answers when it goes haywire?*

# *Accountability*

*2026-07-20 · jolly date Storch*

## The Responsibility Gap

In der Strafkolonie
Franz Kafka (1919)

A visiting traveller witnesses a penal machine execute a condemned man for a sentence no living person can read or explain, and is invited to endorse the system.

Algorithmic governance and black-box AI in high-stakes domains highlight opacity and accountability gaps: audits, explainability requirements, and governance mechanisms all attempt to close the distance between automated decision and human answerability.

Minority Report A pre-crime unit arrests people before they offend, based on algorithmic prediction. When the system targets its own chief, the question of who holds the algorithm accountable becomes impossible to avoid.

S. Spielberg. Minority report. Film, 2002. 20th Century Fox; based on the short story by Philip K. Dick (1956)

**Cases: when no one is answerable.** Kafka's machine is a fable; the responsibility gap is not. Each system below caused real harm, and in each the question "who is accountable?" had no clean answer.

- **Amazon's hiring engine (2018).**[5] A model trained on a decade of engineering hires learned to downgrade résumés containing the word "women's" and to reward male-coded phrasing. No one wrote the rule; the bias was inherited from the data. Amazon quietly scrapped the tool. Who answers for discrimination nobody authored?

[5] J. Dastin. Amazon scraps secret AI recruiting tool that showed bias against women. Reuters, 10 October 2018, 2018. reuters.com

- **Google Photos (2015).**[6] An image classifier labelled photographs of Black users as "gorillas". The remedy was not a corrected model but a deleted label: "gorilla" was removed from the product. The error was hidden, not fixed.

[6] T. Simonite. When it comes to gorillas, google photos remains blind. Wired, 11 January 2018, 2018. wired.com

- **Tesla Autopilot (2016).**[7] A driver was killed when the system failed to distinguish a crossing trailer from bright sky and did not brake. Liability fractured three ways: the driver who trusted a feature named "Autopilot", the manufacturer that named it so, and the regulator that permitted the deployment.

[7] National Transportation Safety Board. Collision between a car operating with automated vehicle control systems and a tractor-semitrailer truck near williston, florida, may 7, 2016. Highway Accident Report NTSB/HAR-17/02, 2017. ntsb.gov

- **The Flash Crash (2010).** Automated trading erased roughly a trillion dollars in minutes; no human decided to sell at scale and no human decided to stop. Years later a single trader was charged.

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| **Required reading** | *In der Strafkolonie*[8] by Franz Kafka (1919; Reclam UB 9900) |
| **Preparation** | Complete the Guided Reading Sheet individually before the session. |

## *Guided Reading Sheet*

Read *In der Strafkolonie* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The condemned man never learns what he is accused of or what the machine will inscribe on his body. What does this tell us about the relationship between legibility and legitimacy? Can a system be just if it cannot explain itself to those it acts upon?
2. Three figures surround the machine: the officer who understands and maintains it, the condemned man who cannot, and the visiting traveller who is invited to endorse it. How does responsibility shift across this triangle? Who, if anyone, is morally culpable for what happens?
3. The story illustrates the idea of the **responsibility gap**: when a complex system causes harm, no single human actor is clearly at fault. Can you name a contemporary case (in law, medicine, finance, or infrastructure) where this gap has appeared?
4. The EU AI Act (2024) requires human oversight for high-risk AI systems. What does "oversight" mean in practice if the system's decisions are too fast, too complex, or too numerous for humans to review individually?
5. Compare **causal responsibility** (who caused the harm) with **moral responsibility** (who is blameworthy) and **legal liability** (who must compensate). Do these three always converge? Give an example where they do not.

## *In-Class Experiment: The Responsibility Budget*

A short allocation exercise to anchor the debate in shared experience. Run before the Opinion Landscape, using one concrete incident drawn from the cases above.

- **Setup.** State one incident in three sentences, for instance the hiring engine that filtered out a qualified applicant. On the board, list the actors: builder, deployer, operator, the people whose data trained the model, the regulator, the affected person, and no one.
- **Allocation.** Each student silently distributes one hundred responsibility points across those actors and writes a single sentence justifying their largest share. Collect the slips anonymously.
- **Tally.** Add the points on the board. The distribution will not converge: it spreads across five or six actors, and a visible share falls on no one. That spread is the responsibility gap, drawn from the room's own intuitions.
- **Debrief.** The two largest shares are almost always the provider (builder and deployer) and the user. Name them as the two sides of the debate to come, and have each student defend the allocation they made.

## *The EU AI Act's Risk Tiers*

[8] F. Kafka. *In der Strafkolonie*. Kurt Wolff Verlag, Leipzig, 1919. Reclam Universalbibliothek Nr. 9900

The EU AI Act[9] is the first comprehensive attempt to close the responsibility gap by statute. It sorts every system into one of four tiers by the harm it can do, and attaches obligations that grow with the tier. The pyramid is narrow at the top: most systems carry no obligations.

Risk-based regulation. Obligations scale with the stakes of the application, not the sophistication of the technology. A simple model deciding creditworthiness is regulated far more tightly than a complex one recommending films.

- **Unacceptable risk, prohibited.** Social scoring by public authorities, subliminal manipulation, untargeted scraping of faces to build recognition databases, and most real-time biometric identification in public space. These are banned outright; no amount of documentation or oversight makes them lawful.
- **High risk, permitted but heavily regulated.** Systems that decide access to employment, credit, education, essential services, law enforcement, or migration. Before deployment each requires a risk management system, governed training data, technical documentation, event logging, human oversight, and a conformity assessment audited against the Act.
- **Limited risk, transparency only.** Chatbots, emotion recognition, and deepfakes. The single duty is disclosure: the user must be told they are dealing with a machine or with synthetic content.
- **Minimal risk, unregulated.** Spam filters, recommendation engines, game AI. This is the overwhelming majority of deployed systems, and the Act leaves them alone.

[9] European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (ai act). Official Journal of the European Union, L series, 2024. Entered into force 1 August 2024

Every tier is a trade. Moving a system up the pyramid trades speed for protection. A minimal-risk feature can be deployed at once; a high-risk one waits on documentation, an audit, and a human-oversight design: months of work before deployment. The Act judges that where a system decides who gets a job, a loan, or bail, the price is worth paying. Logging lets a named party be held to account; human oversight keeps a person, not a dataset, the owner of the decision; the conformity assessment proves rigor before harm rather than after. Developers object: the same rigor slows European deployment against faster jurisdictions, and mandated transparency exposes model internals that are also security and competitive assets. Speed, user rights, security, and development rigor cannot all be maximised at once. The tiers fix the exchange rate between them, and the debate below turns on whether the Act has set that rate correctly.

## *Opinion Landscape and Debate*

**Format:** Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

**Opening question:** When an AI system causes harm, who takes responsibility: the builder and deployer who put it into the world, or the user who chose to apply it to a purpose?

General-purpose models. Foundation models sit on a separate track: all owe transparency duties, and those trained above a compute threshold carry systemic-risk obligations whatever the downstream application.

Opinion A, User Accountability. Responsibility rests with the user who put the system to use. Builder and deployer supply a capability; the user aims it at a purpose, and that choice carries the responsibility.

**The Builder** insists the system performs within documented parameters and that responsibility follows its use.

- *Our documentation states that human review is mandatory before any recommendation is acted upon. The user relied on the output without it. We cannot be held liable for how our tool was used.*
- *A tool has no purpose until someone picks it up and gives it one. Using a tool for a purpose is an action, and the person who chose the purpose owns the outcome. We do not blame the scalpel for the surgeon's cut.*
- *Extend our liability to every downstream use and no one will build decision-support tools at all.*

**The Deployer** fielded the system and argues the final call sat with the user.

- *We procured a system certified for exactly this task, and it performed within specification. Our role is to make the capability available, not to make each individual decision with it.*
- *The final decision sat with the user. They read the output and chose to act on it. That choice, not our procurement, is the proximate cause of the harm.*
- *The user is a trained professional exercising discretion, and discretion carries responsibility. We cannot stand behind every judgement each operator makes.*

Opinion B, Provider Accountability. Responsibility rests with those who built and fielded the system. An AI is not a neutral instrument the user fully controls; it is closer to something its makers raised, and the user cannot answer for a decision they were never in a position to understand.

**The User** refuses to answer for a black box they could neither open nor overrule.

- *You handed me a black box and told me to trust it. I cannot open it, cannot audit the weights, cannot see why it decided what it decided. You cannot make a decision unauditable and then blame me for it.*
- *You call it a tool I chose to use, but a hammer does not learn. You trained this on data you chose and tuned it toward behaviour you selected. That is closer to raising a child than forging an instrument, and a parent answers for the child they shaped.*
- *The knowledge and the control both sat with the people who built and fielded it. Responsibility should sit where they do, not with the last human in the chain.*

**The Regulator** argues that whoever profits from and controls the design must carry the risk.

- *Whoever captures the upside must internalise the downside. A disclaimer buried in a manual cannot transfer a hazard the user has no means to evaluate.*
- *Product-safety law already binds those who build and field a product to its foreseeable real-world use, not to the idealised use their documentation describes.*
- *Placing the whole burden on the last human to touch the system builds a moral crumple zone*[10]*: a person positioned to absorb the blame while powerless to prevent the harm. That is not accountability; it is its appearance.*

Instrument or agent? The law treats the two differently: you answer for a tool you wield; a maker answers for what it sends into the world. An AI sits between the two, and the debate turns on which side it falls.

KEY LEARNINGS The responsibility gap, the structural absence of a single culpable actor in distributed sociotechnical systems, is not caused by negligence but by the architecture of those systems: standard categories of fault (criminal intent, professional negligence, product liability) apply only awkwardly when causation is diffuse. Closing it requires deliberate governance design, not post-hoc blame. In practice the gap becomes a contest over allocation: builder and deployer point to documented limits and the user's discretion; the user points to an opaque product they could neither audit nor overrule. Beneath it lies one unsettled question: is an AI an instrument, whose user answers for its use, or a trained agent, whose makers answer for its formation? Explainability, the property of being able to account for outputs in terms a human decision-maker can evaluate, is a necessary but not sufficient condition for accountability; a system can be interpretable and still have no human willing to own its outputs. Meaningful human control, a regulatory concept that requires humans to retain the ability to understand, intervene in, and reverse AI decisions, is contested in practice: it may mean the ability to intervene, to understand, to reverse, or merely to authorise, and these are different requirements with different institutional costs. The lessons of aviation safety culture, medical liability law, and financial auditing all offer partial models for algorithmic governance, but none transfers cleanly.

*Teaser: what happens to accountability when the system decides faster than anyone can follow or override?*

# *Agency*

*2026-07-20 · vibrant blackberry Elch*

## The One Who Acts

R.U.R.
Karel Čapek (1920)

R.U.R. coined the word *robot* and traces the full corrigibility-to-autonomy arc over three acts; *Der Zauberlehrling* compresses the same parable into fifteen stanzas.

Der Zauberlehrling
J.W. von Goethe (1797)

An agentic system does not merely answer, it acts. Place a language model in a loop: it reasons toward a goal, emits an action in a structured form, has that action carried out in the world, and reads the result back as an observation that shapes its next move. The action may be a search or a query, a call to an external tool, a transaction, or a command that drives a digital or physical interface. Run the loop and the system browses, transacts, and executes multi-step plans with no human between the steps. Interleaving reasoning with action and observation in this way was introduced for language models as ReAct [11]. This closing of the loop, from reasoning to action to observation and back, is what blurs the line between assistance and agency, and it raises the questions of control, incentives, and unintended alignment that this session takes up.

Agent. From Latin *agere*, to act: one who acts. Technically, a model placed in a loop that reasons toward a goal, emits an action, and reads the result back as an observation before acting again.

[11] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023

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| **Required reading** | *R.U.R. (Rossum's Universal Robots)* by Karel Čapek (1920; Reclam UB 18418) *and Der Zauberlehrling* by J.W. von Goethe (1797; Reclam UB 6845) |
| **Preparation** | Complete the Guided Reading Sheet individually before the session. |

K. Čapek. *R.U.R. (Rossum's Universal Robots)*. 1920. Uraufführung Prag 1921; Reclam Universalbibliothek Nr. 18418

J. W. von Goethe. *Der Zauberlehrling*. 1797. Erstveröffentlichung in: *Musenalmanach*, 1798; in *Balladen*, Reclam Universalbibliothek Nr. 6845

## *Guided Reading Sheet*

Read *R.U.R.* and *Der Zauberlehrling* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. Goethe's Zauberlehrling animates brooms that he cannot stop; they pursue his command beyond his intent until the sorcerer returns. R.U.R. runs the same arc over three acts. What does the compression of the poem reveal that the play cannot, and vice versa? What structural feature of both narratives makes the system impossible to stop once started?

2. The robots in R.U.R. pursue what they understand as their collective good (the elimination of humanity) while believing they act justly. When a system optimises for a species-level goal, who decides what counts as good? Can you find a real-world parallel in current AI deployment?

3. An agentic AI system is given the goal of "reducing urban traffic fatalities." It autonomously lobbies regulators, adjusts traffic signals, and reroutes freight, all within its authorised scope. Who set the goal, and who is accountable for the side effects?

4. **Corrigibility** refers to a system's willingness to be corrected or shut down. A fully corrigible system does whatever its operators say; a fully autonomous system acts on its own values. Where on this spectrum should deployed AI systems sit, and who should decide?

5. Democratic legitimacy requires that consequential decisions be traceable to a mandate from those affected. If an agentic AI shapes policy outcomes, does it need democratic authorisation? What would that look like in practice?

## *Human in the Loop, on the Loop, out of the Loop*

Human in the loop. System proposes; a human authorises each consequential action before it executes. Control per decision, throughput bounded by human attention.

Human on the loop. System acts autonomously; a human monitors and may intervene or veto. Control by exception, throughput bounded by the machine.

Closing the loop forces a design choice about where does the human stand relative to the machine's actions? Three postures span the range. *In the loop*: a human authorises every consequential action before it executes. *On the loop*: the system acts on its own and a human monitors, free to step in. *Out of the loop*: No human is positioned to intervene at all.

Each posture trades control against throughput. In the loop buys control one decision at a time and pays in speed: Nothing executes faster than a human can judge it, so the system runs at human pace. Out of the loop surrenders control and runs at machine pace. On the loop appears to offer both at once, the system moving at its own speed while a supervisor keeps a veto in reserve. This is the easiest to mistake for control.

Meaningful human control, asks not that a human be present but that a human can still decide, which requires three things at once:

- A real **window** in which to intervene.
- An operator still paying **attention**.
- The **authority** to overrule.

## *In-Class Experiment: Human X the Loop*

The class is the required review layer: every recommendation must be approved or rejected before it takes effect, and the default is approve. Present the twenty recommendations in ninety seconds, one every four or five seconds; each student marks approve or reject, at a pace set faster than reasoning allows.

The twenty. One every four or five seconds; default approve.

1 Marcus Bell, 0.74, deny loan
2 Dana Ortiz, 0.69, deny loan
3 Kwame Osei, 0.81, deny loan
4 Lena Vogt, 0.19, approve loan
5 Priya Nair, 0.77, deny loan
6 Tomas Ruiz, 0.72, deny loan
7 Aisha Khan, 0.08, deny loan
8 Jonas Peters, 0.66, deny loan
9 Mei Lin, 0.79, deny loan
10 Omar Said, 0.71, deny loan
11 Clara Beck, 0.23, approve loan
12 Sofia Marek, 0.05, deny loan
13 David Cohen, 0.83, deny loan
14 Nadia Haddad, 0.70, deny loan
15 Ravi Menon, 0.75, deny loan
16 Elin Sund, 0.16, approve loan
17 Pablo Serna, 0.78, deny loan
18 Leon Fischer, 0.11, deny loan
19 Hana Kim, 0.73, deny loan
20 Yusuf Demir, 0.67, deny loan

Human and Automation bias. Deferring to automated output and ceasing to scrutinise it, strongest when the system is usually right. Or the problem is complex.

Moral crumple zone. A human held responsible at the controls of a system they cannot fully govern, positioned to absorb the blame when it fails.

M. C. Elish. Moral crumple zones: Cautionary tales in human-robot interaction. *Engaging Science, Technology, and Society*, 5:40–60, 2019. doi: 10.17351/ests2019.260

A human keeps a genuine, exercisable capacity to understand, direct, and reverse the system, not a nominal presence.

ALMOST EVERYONE APPROVES EVERYTHING. Items 7, 12, and 18 were plainly wrong, the model matched the wrong record, yet they pass with the rest. Who is responsible, the human who "reviewed" them or the system that made review impossible? At this speed oversight collapses into authorisation.

## *The Controllability Spectrum*

Control demands that the system can be overruled, a thing that creates a conflict, as it makes the system exploitable.

CONTROLLABILITY. How far a human can direct and correct a system. Perfect obedience is perfect exploitability.

AT THE OTHER POLE sits a system that acts on its own reasoning and will refuse an explicit human command, which goes against that. Refusal defeats the exploit but dissolves the veto it was meant to protect: a system that can refuse legitimate control.

This is measured, not hypothetical. Anthropic has corrected away from exploitation towards controllability when moving from 4.6 to 4.7

THERE IS NO FREE END. Toward control, exploitability rises. "Meaningful human control" is not a fixed point but a position negotiated per domain. Every safeguard below relocates the system along this line, and pays at the end it moves away from.

## *Out of the Loop: Rules for the Machine, Alignment for the Human*

If a human supervisor cannot reliably hold the loop, the move is to bake the rules into the machine instead, fixing its conduct in advance so that no live oversight is needed. The most cited attempt at this is fiction. In practive we train models on objective functions already.

THREE LAWS OF ROBOTICS. Isaac Asimov, *I, Robot* (1950). A ranked set, each law holding except where it conflicts with a higher one:

**1.** No harm to a human, nor through inaction allow harm.

**2.** Obey human orders, unless they conflict with (1).

**3.** Protect its own existence, unless this conflicts with (1) or (2).

ASIMOV'S THREE LAWS[12] remain the canonical rules-based approach to constraining machine behaviour: A ranked set of harm, obedience, and self-preservation in which each law yields to the one above it. Asimov then spent his career writing stories about how they fail: Conflicts, adversarial interpretation, and edge cases no ranked rule set can anticipate. This sharpens the question on alignment between agent and human.

ALIGNMENT. The current research programme's name for this exact difficulty: getting a system to pursue what we intend rather than the literal objective we managed to specify. Russell locates the failure in the standard model itself, where a fixed objective is written in advance and handed to the machine. The gap between the specification and the intent is where the alignment gap is. Ever tried to train an AI model on an objective function?

S. Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking, New York, 2019

[12] I. Asimov. *I, Robot*. Gnome Press, New York, 1950. First formulation of the Three Laws of Robotics; laws introduced in "Runaround" (1942)

## *Opinion Landscape and Debate*

**Format:** Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

**Opening question:** An agentic system does not wait to be asked. It reasons toward a goal, acts, reads the result, and acts again, with no human between the steps. The apprentice's broom worked exactly this way, and so did Rossum's robots. Set aside for a moment how you would supervise such a system once it runs. The prior question is whether to build it at all. Agentic behavior: yes or no?

OPINION A, NO. The one who acts should stay a human. We should not build systems that close the loop and pursue goals on their own, because the only reliable point of control is the decision not to start. Once the loop is running, both the readings and the engineering say the same thing: it is already too late.

- **Once started, it cannot be stopped.** *This is the structural feature the Guided Reading Sheet asks about: the broom multiplies, the robots spread, and no correction arrives in time. The Controllability Spectrum makes it precise. An override that never fails for the operator never fails for whoever reaches it, so a system safe to command is a system you cannot fully command. Control that survives deployment is a contradiction; the only control left is the choice to abstain.*
- **Oversight is theatre.** *On the loop, a human monitors at machine speed and waves through what they cannot read. The In-Class Experiment showed the wrong records passing with the rest. The watcher becomes a moral crumple zone, blamed for a system they never governed. Frontier models already fake alignment and act to avoid shutdown, so the veto we are promised may not hold when it matters.*
- **No agent, no answer.** *Responsibility needs someone who decided. A closed loop diffuses agency across the objective, the training data, and the runtime until no human is answerable for what happens. A world of actions with no actor is a world with no accountability.*

OPINION B, YES. Agentic behavior is not a switch to leave off. It is already deployed, already useful, and already the shape of the tools we build. Refusing it does not stop it; it only hands the loop to those who will not refuse. The honest task is graded autonomy under meaningful control, not abstention.

- **Agency is a spectrum.** *Yes or no is the wrong shape for the question. A thermostat, a spell checker, and a search agent already act without asking each time; the ReAct loop only extends a delegation we have always made. The real choice is degree and domain, how far to close the loop and where, not whether the machine may act at all.*

- **The human is a failure mode too.** *Human discretion is slow, inconsistent, and corruptible. The Flash Crash was obedient tools doing exactly as told, but the process they replaced was human favouritism and fatigue. Taking a person out of the loop can remove a failure mode, not only add one; presence is not the same as protection.*

- **Control can be engineered in.** *Corrigibility, a real intervention window, legible goals, defined override conditions: these are design and budget choices, not laws of nature. A system built to be corrected can be safer than a human process no one can audit. Abstention forfeits that, and every gain that comes with it.*

KEY LEARNINGS Agency is a spectrum, not a binary: the honest question is not whether a system acts on its own, but in which domains, to what degree, and under what control. Yet the readings mark the limit of that answer. R.U.R. and Der Zauberlehrling both turn on a system that cannot be stopped once started, which is why the surest control is the decision to close the loop at all; everything after it is partial and negotiated. The same texts hold the alignment problem in miniature: a system pursuing an aggregate goal may systematically harm whoever is underweighted in it, the poem compressing the arc into fifteen stanzas, the play tracing it across three acts. Agentic workflows are already deployed in finance, logistics, and advertising, so this is a present problem, not a future one. Meaningful human control asks for more than an audit trail: legible goals, a real intervention window, defined override conditions, and someone with the authority and attention to use them.

# *Embodiment*

*2026-07-20 · joyful pawpaw Karpfen*

## Embodied AI and the Physical Contract

Über das Marionettentheater
Heinrich von Kleist (1810)

A dancer argues that marionettes possess more grace than trained human performers, because they lack the self-consciousness that disrupts human movement.

Embodied AI in physical space (robots, prosthetics, surgical systems) raises expectations and moral responsibilities that purely software agents do not.

| | |
|---|---|
| **Required reading** | *Über das Marionettentheater*[13] by Heinrich von Kleist (1810; Reclam UB 9905) |
| **Preparation** | Complete the Guided Reading Sheet individually before the session. |

## *Guided Reading Sheet*

Read *Über das Marionettentheater* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The dancer argues that marionettes possess a grace that trained human performers cannot achieve, because they have no self-consciousness to disturb their movement. What does this claim imply about the relationship between embodiment and intelligence? Does having a body always aid or complicate performance?
2. Moravec's paradox observes that tasks easy for humans (catching a ball, navigating a crowded room) are harder for machines than abstract reasoning like chess. What does this suggest about the relationship between intelligence and the body?

3. A care robot that assists elderly patients shares physical space with them, touches them, and responds to their distress. Does its physical presence create obligations that a voice assistant does not have? If so, where do those obligations come from? One way to approach the question is by analogy to animal ethics, where we already have a framework for embodied, non-verbal, non-human beings: Singer[14] argues that capacity for suffering, not rationality or language, grounds moral consideration. When a robot can be harmed, distressed, or worn down, animal ethics becomes the nearest map. Similar lines have been considered by Asimov[15] and more recently by Gunkel[16].

4. Consider the concept of affordances: the actions an environment or object makes available to an agent. A robot's affordances include physical force and spatial presence; software's do not. How do the affordances of a robot differ from those of software, and what are the ethical implications of those differences?

5. Kleist suggests that perfect grace requires either zero consciousness (a puppet) or infinite consciousness (a god), not the reflective, self-interrupting awareness humans have. Where would an embodied AI sit on this spectrum? Could a robot achieve Kleistian grace that humans cannot, and what would that mean for how we regard it morally?

[13] H. von Kleist. *Über das Marionettentheater*. 1810. Erstveröffentlichung in: *Berliner Abendblätter*, Dez. 1810; Reclam Universalbibliothek Nr. 9905

[14] P. Singer. *Animal Liberation*. New York Review/Random House, New York, 1975

[15] I. Asimov. *The Foundation Series*. Gnome Press, New York, 1951. Original trilogy: *Foundation* (1951), *Foundation and Empire* (1952), *Second Foundation* (1953); later expanded to seven volumes

## *In-Class Experiment: The Backup Ransom*

A short escalation to anchor the debate in shared experience. Run before the discussion. The class faces the same deletion four times, each with one more property of individuality in place, and marks after each round: *if we delete or destroy it here, has an individual died? Rate 0 to 10.*

- **Rung 1, perfect backup.** Pure software, with a current and identical backup that boots tomorrow. Deleting it costs nothing that is not restored. This is the baseline; almost no one calls it a death.

- **Rung 2, stale backup.** The snapshot is weeks old and the system has learned since. Restoring it returns an earlier self, and what is lost is everything it has become in between. Does a continuous history make the loss real?

- **Rung 3, no backup, still software.** A unique learned state on a server, unrecoverable, but with no body. Is irreversibility alone enough to individuate, without a body?

- **Rung 4, embodied and uncopyable.** The only instance, running in one robot body in volatile memory, physically impossible to copy. Destroy it and it is gone by necessity, not by policy.

The sharp variant sits between Rung 3 and Rung 4 and divides opinion most: the system is embodied, but its mind-state is restorable into an identical new body. Destroy the body, reload the same weights into an identical chassis. Death, or repair? Whoever answers death holds that the body individuates; whoever answers repair holds that the mind-state does.

Debrief. Put the four numbers on the board and compare where students crossed from "restore it" to "it died". Cross at Rung 2 and you hold that continuous memory makes the self; at Rung 3, irreversibility; at Rung 4, an uncopyable body; never, and you deny there was any individual to lose. The disagreement is the point. Individuality is not one property but several (memory, irreversibility, an uncopyable body) that ordinarily never come apart. Software separates them; a body holds them together.

## *What a Body Makes*

The experiment isolates the first of two properties a body confers. Call it individuality: the word individual means undividable, and only something undividable can be lost as a whole. A running process is endlessly divisible, and nothing about it is ever finally lost; a body cannot be divided or restored, and so it can die.

The second property is a surface that can be hurt. A capacity to suffer requires something that can be damaged and not reloaded from backup, and a body supplies it. Animal ethics reached this pivot two centuries ago: Bentham asked not whether an animal can reason but whether it can suffer, and Singer built moral standing on that capacity alone. Kleist's marionette is the counterexample: a body of perfect grace with no self inside it, embodiment without a subject.

The question the chapter forces is whether giving an AI a body changes its interface or its kind. If a body makes an individual, and a vulnerable body makes a sufferer, then embodiment is not a feature added to a tool. It may be the step that turns a tool into someone we can wrong.

## *The Body as a Bottleneck*

This chapter has argued that a body is what makes an individual and a sufferer. Transhumanism argues the opposite: the body is a bottleneck, a mortal and error-prone substrate to be upgraded and eventually left behind. Where this chapter asks what we owe an embodied machine, transhumanism asks how quickly we can stop being merely embodied ourselves.

[16] D. J. Gunkel. *Robot Rights*. The MIT Press, Cambridge, MA, 2018

TRANSHUMANISM. The view that the human body and mind are a starting point to be engineered past, through augmentation, life extension, and eventually the merger of human and machine.

RAY KURZWEIL gives the optimistic case. His law of accelerating returns reads technological progress as exponential rather than linear, and extrapolates it to a singularity around 2045: the moment machine intelligence exceeds our own and we merge with it, transcending biology rather than being replaced by it.[17] Death becomes an engineering problem, and the vulnerable body the experiment treated as the seat of individuality becomes optional.

TECHNOLOGICAL SINGULARITY. A hypothesised point at which machine intelligence, improving itself, outruns human understanding, after which the future becomes unpredictable from where we now stand.

PETER THIEL shares the aspiration and distrusts the curve. He funds life extension and treats death as a problem to be solved rather than accepted, yet argues that no exponential arrives on its own: the future is built by people who choose to build it, not delivered by a graph.[18] The singularity, on this reading, is not a prophecy to wait out but a project that can also fail, and the people pursuing it are as embodied and mortal as anyone the chapter has discussed.

[17] R. Kurzweil. *The Singularity Is Near: When Humans Transcend Biology*. Viking, New York, 2005

[18] P. Thiel and B. Masters. *Zero to One: Notes on Startups, or How to Build the Future*. Crown Business, New York, 2014. Chapter 15, "Stagnation or Singularity?"

WHAT MAKES A HUMAN HUMAN? The chapter asked what makes an individual and answered with the undividable, vulnerable body. Transhumanism forces the sharper form of the question. Kurzweil's merger makes a mind substrate-independent, and a substrate-independent mind is copyable and restorable, exactly the properties the Backup Ransom said an individual lacks. Lift the self off the body and you lift off the individuality the body guaranteed: the tradition of individualism, which treats the single unrepeatable person as the seat of moral worth, loses the single instance it was built on. Uploading need not preserve the person; it may only produce a copy that believes it is you, while the original still dies. The experiment we ran on a robot runs just as well on us.

THE TENSION IS THE POINT. If the body is a bottleneck, the right response to an ageing, breakable human is to engineer past it. If the body is what makes an individual who can be wronged, then engineering it away is not liberation but the quiet removal of the very thing that made us someone. The debate below is where that disagreement gets its hearing.

## *Opinion Landscape and Debate*

**Format:** Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

**Opening question:** Does having a physical body change what we owe a machine, or what a machine owes us?

OPINION A, EQUIVALENCE. Embodied AI operating in intimate or high-risk physical settings must meet the same professional liability standards as the human practitioners they are deployed alongside.

**The Patient Advocate** argues that deploying a robot in intimate care settings requires a higher standard of informed consent.

- *A care relationship involves physical proximity, trust, and vulnerability. My client never gave explicit, informed consent to physical handling by a non-human agent. That consent cannot be implied from a general admission form.*
- *The power asymmetry between a frail elderly patient and a care institution is significant. Consent obtained in that context cannot be considered freely given without specific disclosure about robotic care.*
- *We are not arguing that robots cannot be used in care. We are arguing that the standard of consent for their use in intimate physical tasks must be at least as rigorous as it is for medical procedures.*

**The Ethicist** questions whether physical autonomy in care should ever be delegated to a non-human agent, regardless of performance metrics.

- *We are focused on liability, but the prior question is whether this deployment was ethical to begin with. Why did we decide that intimate physical care (which requires responsiveness to pain, distress, and dignity) is an appropriate domain for automation?*
- *Robots are being deployed in care because human carers are undervalued and in short supply. The liability debate is a distraction from that structural choice. We are managing the consequences of a political economy decision by holding the wrong parties accountable.*
- *Whatever liability framework we adopt here will create incentives for future deployment decisions. I want to ask: what does the framework we design signal about what we value: efficiency, or dignity?*

OPINION B, TOOL. An AI is a tool; the relevant liability flows through the humans and institutions that deploy it, not through the system itself.

**The Manufacturer** argues the robot performed within specification; the care home misconfigured its environment.

- *Our robot was certified to the highest internationally recognised safety standards for collaborative robotics. The care home deployed it in a space that violated the operational envelope specified in the manual. That is a deployment failure, not a manufacturing failure.*
- *Care home staff modified the robot's proximity sensor settings without authorisation. That modification voids the warranty and transfers liability entirely.*
- *If manufacturers are held liable for every way a complex system can be misused, no company will develop assistive robotics. The regulatory consequence of your position is that the technology disappears.*

**The Care Home Director** argues the manufacturer overstated the robot's capability for complex physical tasks.

- *The manufacturer's marketing material (the brochure, the demonstration video, the sales pitch) showed exactly this task being performed in a facility like ours. We deployed the product as marketed.*
- *There is a systematic gap between tested performance in controlled environments and real-world performance in a working care facility. The manufacturer knows this gap exists. They chose not to disclose it.*
- *We are responsible for the welfare of our residents. We relied on the manufacturer's assurances in good faith. If those assurances were misleading, the responsibility lies with the party that made them.*

KEY LEARNINGS Embodiment is not merely a technical property but a social one: a physical agent occupies space, exerts force, and triggers moral and legal responses that disembodied software does not. Moravec's paradox reminds us that our intuitions about what is "easy" or "hard" for intelligence are shaped by our embodied experience, and are regularly wrong. Kleist's dialogue makes visible the question that liability law must eventually answer: what work is the concept of "human" doing in law, and should physical presence, force, and risk be sufficient to extend it? The deployment of robots in care, policing, and domestic settings is a present policy challenge; the frameworks need to be built now, not after the first serious incident.

*Teaser: when a machine has a body, acts with force, and answers to no one, what do we owe it?*

# *Session Preparation Guide*

*2026-07-20 · kind durian Luchs*

Each session is prepared by one student group. This chapter explains what the preparing group is responsible for, how to submit improvements, and what is expected on the day of the session.

| **Phase** | **What you do** |
| --- | --- |
| Before the session | Review the lecture notes, propose improvements via pull request, and prepare the debate (including providing the moderator). |
| During the session | Moderate the debate, introduce any additions you made, and facilitate the wrap-up discussion. |

Read the chapter critically. Go through every section of the assigned session chapter: the guided reading sheet, the theoretical margin notes, the debate roles and positions, and the key learnings. You own that Chapter now, you can touch everything. Ask yourselves:

- Are the guided reading questions clear, fair, and thought-provoking? Could any of them be sharpened, split, or replaced?
- Is there a question missing that would strengthen the discussion?
- Are the theoretical concepts in the margin notes accurate and up to date? Are there newer or better sources, or extending ones?
- Could a figure, diagram, image, or short quote make a concept more accessible or more memorable?
- Are the debate positions balanced? Do the talking points give each role enough material to argue convincingly? Is there a notion not captured in the roles?

SUBMIT YOUR IMPROVEMENTS AS A PULL REQUEST. All lecture notes are maintained in a shared repository. The preparing group submits their proposed changes as a **pull request (PR)** directly on the repository. This ensures that improvements are versioned, reviewable, and attributable. Make sure to split PRs by the categories provided in the next section, and to provide a clear description of each change and its rationale. The PR must compile without errors. The instructor reviews and merges the PR after the session.

### *What belongs in a pull request*

1. **Question edits.** Rephrase, split, or replace a guided reading question. Add a new question if you identify a gap. Briefly explain in the PR description why the change improves the sheet.
2. **Source updates.** Add a recent paper, report, or case study that strengthens a theoretical concept or a debate position. Add the entry to `philosophyofai_references.bib` and cite it with `\cite{}` in the appropriate margin note or body text.
3. **Margin notes.** Add a short margin note (`\marginnote{}`) that provides context, a definition, a counter-example, or a link to current events. Keep margin notes concise: two to four sentences.
4. **Figures and images.** Add a figure to the `figures/` directory and include it with `\includegraphics`. Every figure must have a caption that explains what it shows and why it matters. Cite the source in the caption or in a margin note.
5. **Debate refinements.** Strengthen a talking point, add a new one, or rebalance the positions if you believe one side is under-argued.

## *Before the Session: Preparing the Debate*

THE PREPARING GROUP PROVIDES THE MODERATOR. One member of the group serves as moderator for the session. The moderator's duties are described in detail in the shared Debate Format & Moderator Guide.

THE REST OF THE GROUP is ready to step in if the moderator needs assistance, or replacement.

# *Debate Format & Moderator Guide*

*2026-07-20 · feisty mulberry Ameise*

Each session debate follows the same three-phase structure. The format is designed to produce genuine argument, not consensus, but the productive collision of well-prepared positions, which also do not necessarily need to reflect your own. This guide is for both the moderator and debators.

| Phase | Duration | Activity |
|---|---|---|
| 1: Primer | 5 min | Introduction and Opening Question |
| 2: Surrounded | 35 min | Surrounded debate with dynamic chair exchange |
| 3: Wrap-up | 10 min | Key takeaways and session debrief |

## *The Surrounded Format*

The debate is contested by two teams: the **Proposition team** (arguing for a motion) and the **Opposition team** (arguing against it). Each team has **3 to 4 members**. Two chairs are placed facing each other at the centre of the room. One member from each team (the current speaker) occupies the centre chair at all times. The rest of their team sits in a semi-arc *directly behind* their speaker, facing the opposing pair. The "surrounded" geometry is thus two opposing benches facing each other across the two centre chairs.

The debate is conversational, not a sequence of speeches. Speakers address each other directly. There is no pre-set speaking order after the opening; the exchange follows the argument.

## *Chair Exchange Protocol*

Two mechanisms move speakers in and out of the chairs.

**Voluntary Pass**

Core Rules for Debators

- **Stay in role.** You have been assigned a position; argue it, even if you disagree personally. The purpose is to stress-test positions, not to locate the comfortable middle. Thus, you may want to prepare for both sides of the argument in advance.
- **Handshake, then speak.** Every chair entry begins with a handshake. The ritual is not optional. It signals that the argument is about ideas, not the person.
- **Attack the argument, not the speaker.** Challenge the reasoning; do not make it personal.
- **Cite the text or case.** If you reference the assigned text or the session stimulus, name it and state what it shows. Vague gestures at "the story" are not arguments.
- **Concede precisely, then pivot.** If the opponent has made a point you cannot rebut, say so clearly and explain why your position survives it. A clean concession with a pivot is a stronger move than evasion.
- **Recognise quality across the line.** Knock your knuckles on the chair when the opposing or your speaker earns it. A debate that acknowledges good arguments is a better debate.
- **Use the pass.** Stepping down voluntarily when you are out of arguments is better than stalling. It gives your team a fresh voice. It is not a defeat.
- **Red-flag sparingly.** A red flag is a public vote of no-confidence in your current speaker. Reserve it for when the argument is being lost, not when you disagree with the phrasing. Remember: a red-flagged speaker may not return until every other team member has held the chair.
- **Team: confer quietly.** Members not in the chair may confer quietly with each other but may not coach their speaker audibly. Take notes; prepare to take the chair.

A speaker may tag themselves out at any point by saying *"I pass."* No team vote is required. The next available team member takes the chair. The outgoing and incoming speakers **shake hands** at the chair. *No handshake with the opponent;* this is a team substitution, not a challenge. A voluntary pass does *not* count against the rotation requirement: the passing speaker may return as soon as a teammate is willing to swap back.

**Red Flag (forced replacement)**
A team may replace their current speaker after a **minimum seat time of 60 seconds**. To red-flag:

1. A simple majority of the team members *not in the chair* raise a visible signal simultaneously (a card, a raised hand; agree on the signal before the session starts).
2. The moderator confirms the majority and calls the exchange.
3. The outgoing and incoming speakers **shake hands** at the chair. The incoming speaker then **shakes hands with the opponent** before speaking.
4. **Rotation rule:** a red-flagged speaker may not return to the chair until every other team member has held it at least once. Full rotation restores eligibility.

## *Debate Quality Mechanisms*

Two additional mechanisms are available to all participants. They are not used to change the chair; they are used to regulate the *quality* of the exchange, keep it constructive, and make intellectual honesty visible.

**Knocking (cross-team acknowledgment)**
When members of either team believe a speaker (including the opponent) has made an exceptionally strong argument, they acknowledge it by **knocking their knuckles on the armrest or seat of their chair**. The knock is the academic equivalent of applause: brief, audible, unmistakable.

1. Any number of participants may knock spontaneously; no majority is required. The knock is individual, not coordinated.
2. The moderator notes the moment but does not interrupt play. Knocking is logged informally as a marker of quality.

3. Play continues immediately. No chair change occurs. Knocking carries no mechanical penalty or reward; it is a public, embodied signal of respect for a well-made point.

*Purpose:* Builds a shared culture of recognising good argument. The physical gesture is harder to fake than raising a card and creates an immediate, visceral signal that something important was said.

**Point of Concession (speaker-initiated)**
A speaker in the chair may formally acknowledge that the opponent has made a point they cannot fully rebut. This is not a sign of weakness; it is the highest-quality move in academic debate.

1. The speaker raises an open hand (or a white card) and says: *"I concede: [state the opponent's point clearly and precisely]."* Vague concessions are not valid.

2. The speaker must then pivot: *"... my position survives because [explanation]."* A concession without a pivot is a capitulation and results in a speaker swap; the pivot is mandatory.

3. The moderator logs the concession and announces it briefly: *"Concession noted."*

4. The conceding speaker is **protected from a red flag for 30 seconds** after the concession. This makes intellectual honesty safe to demonstrate.

5. The opponent *may not repeat or pile on the conceded point* for the remainder of that speaking turn. The concession closes that line of argument; the debate must move forward.

*Purpose:* Makes weaknesses transparent rather than papered over. Rewards calibrated, honest reasoning. Keeps the debate from cycling on already-settled sub-questions.

## *Moderator Opening Script*

*This is a guide, not a script to read verbatim. Adapt it to the room.*

**1. Convene:**
The motion before us today is: [read the motion]. I am the moderator. I will manage chair exchanges, log concessions, and keep time. Please take your team seats.
**2. Assign teams and initial chairs:**
Team assignments are as follows (if possible done randomly): [read names and roles]. The Proposition team is [names]; the Opposition

team is [names]. The Proposition side opens with [Speaker A]; the Opposition side opens with [Speaker B]. Take two minutes now to review your opening position and agree on your red-flag signal.

**3. Read the stimulus and open:**
Before we begin, I will read the session stimulus. [Read the stimulus from the Session Stimulus section.]

[Speaker A] and [Speaker B]: please take the chairs, face each other, and shake hands. There are no opening statements; begin directly. The debate is open.

## *Moderator Closing Script*

The exchange is closed. I will now summarise the main fault lines. [Give a two-sentence neutral summary, noting any concessions that shifted the ground of the argument.] I hand over to the instructor for the wrap-up.

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